

Supplementary material

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British University Vietnam, Hung Yen, Vietnam. Supplement to “Light-sharing control for tracker-based agrivoltaics under a yield floor”.

Abstract

This document contains sensitivity analyses and tables that support the main text: robustness of the controllers to the floor, forecast errors, layout and rotation limit, the statistical comparison of the controllers, the cross-validation and curves of the shade-response model, and the economic sensitivity analysis. Section and table numbers with the prefix S refer to this document; other numbers refer to the main text.

S1. Floor conditioning, chance-constrained training and forecast errors

One network for all floors. Regulations and farmers set different floors. The network of the main experiments was trained at $\rho = 0.9$ only. We evaluated it at floors of 0.80, 0.85, 0.90 and 0.95, re-running the historical design step for each floor (Table S1). Its LER exceeds that of the phenology rule by 0.037, 0.036, 0.034 and 0.033 under the baseline climate (0.040, 0.037, 0.032 and 0.033 at +3 °C), so the benefit carries over to other floors without retraining. A network that received the floor as an additional input and was trained over all four floors did not improve on it (differences in LER ≤ 0.001). Because the design step already adapts the density to the floor, conditioning the controller on the floor is not needed.

Layout, rotation limit and harvest-index assumptions. We checked three assumptions that could drive the ranking (Table S2; 21 pairs of the main split, one training seed). The rotation limit of the trackers sets how much light sharing can deliver: with a limit of 45° the LER of the phenology rule is 1.138, with 60° 1.170 and with 80° 1.206, whereas static trackers stay at 1.05–1.06, so the benefit of light sharing grows with the limit and the ranking is unchanged. Fixed non-tracking arrays held flat or rotated by

Table S1: LER of the controllers at different yield-retention floors ρ (main split; the historical design step is repeated for each floor).

Method	Baseline climate				+3 °C			
	0.80	0.85	0.90	0.95	0.80	0.85	0.90	0.95
AV static	1.159	1.115	1.072	1.011	1.174	1.128	1.083	1.013
Stress rule	1.380	1.282	1.182	1.077	1.364	1.269	1.172	1.071
Phenology rule	1.467	1.356	1.236	1.108	1.513	1.395	1.265	1.122
SALS (trained at $\rho = 0.9$)	1.505	1.392	1.270	1.141	1.553	1.432	1.297	1.154
SALS conditioned on ρ	1.504	1.391	1.269	1.141	1.553	1.431	1.296	1.154

30° about the north–south axis (facing east, not the equator; equator-facing
tilts are not modelled) do no better than static trackers (LER 1.048 and
1.043 against 1.053; differences of 0.005 are within the noise of 21 pairs and
one training seed), so static trackers are no worse than the fixed arrays we
tested; vertical bifacial layouts are not modelled. The ranking of the rules
also does not depend on the assumed harvest-index penalty. The window test
is weakly informative about the timing of the phenology rule, which shares
light through almost the whole growing period, so a shifted window cannot
move it much; the informative case is the removal of the penalty. Moving the
critical window to 0.30–0.50 or 0.60–0.80 of the thermal-time requirement
changes the LER of the phenology rule by at most 0.001, and removing the
penalty altogether raises all light-sharing LERs (phenology rule 1.328, SALS
1.353) without changing the order: the phenology rule exceeds seasonal shar-
ing by 0.03 to 0.05 in every case (0.035 with the early window, 0.034 with
the late window, 0.053 without penalty), because it produces electricity be-
fore canopy development and after maturity, when the crop does not use the
light, and not because of the assumed window. The margin of SALS over the
written rule changes with these assumptions (0.005 to 0.025 LER), and its
margin over the tuned rule is 0.000 to 0.009 (at a rotation limit of 45° SALS
equals the tuned rule).

A floor of 66%. The German pre-standard DIN SPEC 91434 requires
66% of the reference yield (DIN, 2021). Repeating the design step with
 $\rho = 0.66$ (baseline climate, main split, same network trained at 0.9) the
ground coverage cap of 0.5 binds for the sharing methods (mean GCR 0.48
for the phenology rule and 0.48 for SALS, 0.44 for the stress rule), static
arrays reach a coverage of 0.24 and LER 1.28, the phenology rule 1.66 and

Table S2: Sensitivity of the controller comparison to the rotation limit of the trackers and to the harvest-index penalty (LER; main split, every sixth held-out pair, 21 pairs, main shade response, history-based design on a GCR grid with step 0.05). SALS was trained for the main setting. Fixed arrays (no tracking; flat and tilted by 30° from the horizontal about the north–south axis) reach LER 1.048 and 1.043 in the same sample, against 1.053 for static trackers.

Setting	Static tracker	Phenology rule	Grid-tuned window	SALS
Main (limit 60°, window 0.45–0.65)	1.053	1.170	1.180	1.189
Rotation limit 45°	1.057	1.138	1.144	1.144
Rotation limit 80°	1.053	1.206	1.219	1.224
Window 0.30–0.50	1.053	1.169	1.176	1.185
Window 0.60–0.80	1.053	1.171	1.181	1.182
No harvest-index penalty	1.109	1.328	1.347	1.353

51 SALS 1.71. At this floor the density is limited by the cap and by the mean
52 retention of 0.73 reached at the cap, not by the floor, so the comparison of
53 controllers says little about control; we do not interpret it further.

54 **Chance-constrained training.** We also trained a network whose ob-
55 jective penalises a batch share of seasons below the floor that exceeds 20%.
56 It did not improve the tail. With the mean-based design step its LER was
57 1.263 (baseline) and 1.290 (+3 °C) against 1.270 and 1.297, and with chance-
58 constrained design it reached season-level compliance of 92.6% against 92.0%
59 and an LER of 1.198 against 1.205. The season-level guarantee is obtained
60 by the design step (Section 3.1), not by the training objective.

61 **Forecast errors.** We raised the forecast errors seen by the controllers
62 to 0, 1, 2 and 3 times the errors used in training. The LER of SALS and
63 of the phenology rule was identical at all four error levels to three decimals
64 (densities 0.2 and 0.3: SALS 1.307 and 1.457, phenology rule 1.278 and
65 1.416). Neither controller uses the weather forecast: both rely on the crop’s
66 thermal time, which is observable from on-farm temperature records. The
67 stress rule, which depends on the forecast of maximum temperature, degrades
68 with forecast error: at density 0.3 its retention falls from 0.778 to 0.751 and
69 its LER from 1.432 to 1.418 as the errors triple.

Table S3: Paired comparison of SALS with each baseline on the 92 held-out sites of the main split (unit of analysis: site mean over test seasons). ΔLER : mean difference (SALS minus baseline) with cluster-bootstrap 95% CI; p : two-sided Wilcoxon signed-rank test; % better: share of sites where SALS has the higher LER; Δr : difference in mean yield retention with its Wilcoxon p -value.

Climate	Baseline	ΔLER [95% CI]	p	% better	Δr	p	Sites
Baseline	AV static	+0.205 [+0.176, +0.236]	$< 10^{-4}$	100	-0.011	$< 10^{-4}$	92
Baseline	Seasonal sharing	+0.088 [+0.068, +0.109]	$< 10^{-4}$	97	+0.001	0.922	92
Baseline	Phenology rule	+0.038 [+0.025, +0.053]	$< 10^{-4}$	84	+0.001	0.723	92
Baseline	Stress rule	+0.096 [+0.071, +0.124]	$< 10^{-4}$	93	-0.002	0.055	92
+1.5 °C	AV static	+0.208 [+0.180, +0.237]	$< 10^{-4}$	100	-0.010	$< 10^{-4}$	92
+1.5 °C	Seasonal sharing	+0.095 [+0.077, +0.114]	$< 10^{-4}$	98	+0.001	0.812	92
+1.5 °C	Phenology rule	+0.033 [+0.022, +0.045]	$< 10^{-4}$	86	+0.002	0.855	92
+1.5 °C	Stress rule	+0.108 [+0.083, +0.134]	$< 10^{-4}$	98	-0.001	0.041	92
+2 °C	AV static	+0.209 [+0.181, +0.238]	$< 10^{-4}$	100	-0.010	$< 10^{-4}$	92
+2 °C	Seasonal sharing	+0.097 [+0.079, +0.115]	$< 10^{-4}$	97	+0.001	0.876	92
+2 °C	Phenology rule	+0.032 [+0.022, +0.044]	$< 10^{-4}$	85	+0.002	0.496	92
+2 °C	Stress rule	+0.115 [+0.089, +0.142]	$< 10^{-4}$	98	-0.003	0.016	92
+3 °C	AV static	+0.219 [+0.190, +0.250]	$< 10^{-4}$	100	-0.009	$< 10^{-4}$	92
+3 °C	Seasonal sharing	+0.108 [+0.089, +0.129]	$< 10^{-4}$	96	+0.001	0.935	92
+3 °C	Phenology rule	+0.034 [+0.022, +0.046]	$< 10^{-4}$	86	+0.002	0.330	92
+3 °C	Stress rule	+0.131 [+0.102, +0.162]	$< 10^{-4}$	98	-0.002	0.142	92

S2. Statistical comparison

S3. Shade-response model, ensembles and economics

References

- DIN, 2021. DIN SPEC 91434:2021-05: Agri-photovoltaic systems – requirements for primary agricultural use. Beuth Verlag, Berlin.
- Laub, M., Pataczek, L., Feuerbacher, A., Zikeli, S., Högy, P., 2022. Contrasting yield responses at varying levels of shade suggest different suitability of crops for dual land-use systems: a meta-analysis. *Agronomy for Sustainable Development* 42, 51. doi:10.1007/s13593-022-00783-7.

Table S4: Predicting relative yield under shade from 428 observations of 58 studies (Laub et al., 2022): leave-one-study-out cross-validation (every study held out once; observations with $s > 0$). Covariates: irradiation, climate zone, shade type, intercropping.

Model	RMSE	MAE	Median AE	Mean MAE per study
Proportional loss, $y = 1 - s$	0.430	0.305	0.210	0.224
One curve for all crops	0.372	0.265	0.193	0.208
Curve per crop type, no pooling	0.313	0.208	0.141	0.165
Curve per crop type, partial pooling	0.312	0.210	0.150	0.162
Partial pooling + covariates	0.311	0.206	0.148	0.161
Gradient-boosted trees	0.321	0.219	0.162	0.184

Table S5: Leave-one-study-out prediction error (RMSE of relative yield, observations with $s > 0$) by crop type: proportional loss, hierarchical curve with partial pooling, and gradient-boosted trees. The last columns give the difference hierarchical minus proportional with a 95% cluster-bootstrap interval over studies (negative: hierarchical better).

Group	Obs./studies	Proportional	Hierarchical	Boosted t
C3 cereals	30/10	0.214	0.187	0.205
Maize	40/10	0.225	0.235	0.248
Grain legumes	58/14	0.215	0.225	0.234
Tubers/root crops	10/2	0.250	0.254	0.257
Four crop types used by the simulator	138/30	0.220	0.222	0.234
Other types (berries, forages, fruits, vegetables)	201/29	0.528	0.361	0.369
All types	339/58	0.430	0.312	0.321

Table S6: Sensitivity of the deployable light-sharing results to the crop shade response (rule controllers, baseline climate, floor $\rho = 0.9$, split 0, every fourth held-out site, 22 sites). Conservative: proportional response (main analysis); calibrated: radiation-use compensation fitted to Laub et al. (2022); field-fitted: fitted to the field trials, maize harvest index unaffected.

Shade response	Method	GCR	E/E_{PV}	Retention	LER
conservative	AV static	0.06	0.15	0.92	1.07
	Phenology rule	0.17	0.33	0.90	1.23
	Seasonal sharing	0.16	0.28	0.90	1.18
	Stress rule	0.11	0.24	0.91	1.15
calibrated	AV static	0.07	0.18	0.91	1.09
	Phenology rule	0.19	0.37	0.90	1.28
	Seasonal sharing	0.19	0.32	0.90	1.22
	Stress rule	0.13	0.29	0.91	1.20
field-fitted	AV static	0.16	0.41	0.91	1.32
	Phenology rule	0.37	0.70	0.92	1.62
	Seasonal sharing	0.37	0.61	0.92	1.53
	Stress rule	0.29	0.63	0.90	1.53

Table S7: Shade-response parameters of ensemble E2, calibrated by gradient descent through the simulator to the data-driven curves and the field trials (12 randomised fits; mean $\pm$ s.d.): radiation-use compensation c and multiplier h on the flowering-window harvest-index penalty (1 = assumed value).

Crop	c	h multiplier
Wheat	0.91 ± 0.31	0.74 ± 0.53
Rice	0.61 ± 0.25	0.83 ± 0.33
Maize	0.14 ± 0.11	0.16 ± 0.13
Soybean	0.05 ± 0.09	1.66 ± 0.38
Potato	0.62 ± 0.20	0.90 ± 0.50

Table S8: One-at-a-time sensitivity of the net value (kUSD ha⁻¹ yr⁻¹, relative to the open field) at 90 USD MWh⁻¹, densities designed for the floor as in the main analysis. In the capex rows the capex of the conventional plant is set to 0.75 times that of the agrivoltaic plant.

Parameter	Value	Static arrays	Optimised window	SALS	Conventional plant
Base		6.3	7.3	7.6	55.2
AV capex (USD/Wp)	0.7	9.7	16.6	17.1	72.8
AV capex (USD/Wp)	1.5	0.6	-8.2	-8.2	25.8
Discount rate	4%	8.0	12.0	12.4	64.1
Discount rate	8%	4.4	2.1	2.4	45.4
Crop prices	0.5×	6.4	7.4	7.7	57.0
Crop prices	2.0×	6.0	7.0	7.3	51.6
Electricity price	60	0.3	-5.6	-5.6	16.0
Electricity price	120	12.3	20.1	20.7	94.4

Table S9: Design rules outside the ensemble E2 by crop: share of crop–site pairs (%) with mean retention ≥ 0.9 under the proportional (P), calibrated (C), field-fitted (F) and harsh (H) responses (phenology rule, three splits pooled). The number of pairs is given in parentheses; with 10 to 20 pairs per crop a difference of one pair is 5 to 10 points. Pooled values are in Table 8.

Crop (pairs)	Design	P	C	F	H
Wheat (17)	oracle	100	94	100	94
	mean response	0	0	100	0
	75% of members	0	0	100	0
	floor 0.92	0	0	100	0
	floor 0.96	12	76	100	0
Rice (20)	oracle	90	90	80	85
	mean response	0	0	10	0
	75% of members	0	0	40	0
	floor 0.92	0	10	60	0
	floor 0.96	95	95	100	85
Maize (17)	oracle	76	76	71	65
	mean response	6	6	71	0
	75% of members	12	12	82	0
	floor 0.92	18	18	82	0
	floor 0.96	71	71	94	53
Soybean (10)	oracle	90	90	90	90
	mean response	90	90	90	90
	75% of members	100	100	100	90
	floor 0.92	100	100	100	90
	floor 0.96	100	100	100	100
Potato (12)	oracle	100	100	100	92
	mean response	0	0	100	0
	75% of members	0	0	100	0
	floor 0.92	0	0	100	0
	floor 0.96	42	42	100	42